

KU5DSCCSE304 SOFTWARE ENGINEERING

Semester	Course Type	Course Level	Course Code	Credits	Total Hours
5	DSC	300	KU5DSCCSE304	4	60

Learning Approach (Hours/ Week)			Marks Distribution			Duration of ESE (Hours)
Lecture	Practical/ Internship	Tutorial	CE	ESE	Total	
4	0	1	50	50	100	2

Course Description:

This course provides an overview of Software Engineering. It provides the fundamental aspects of the topic and covers the various process models available for creating software projects. It provides insight into the requirement engineering and discusses various techniques for modeling requirements. Fundamentals of software design and component level design in particular is also covered. The course also provides insight into the principles of user interface design, project estimation, scheduling, and testing strategies.

Course Objectives:

- Introduce the basic concepts of software engineering.
- Give basic idea of various software process models.

- Familiarize various processes of software development.

Course Outcomes:

At the end of the Course, the Student will be able to:

SL #	Course Outcomes
CO1	Understand what software engineering is and its importance, characteristics of software, and challenges in the field.
CO2	Explore various software development life cycle models and the principles of agile methodologies.
CO3	Learn techniques for requirements gathering, analysis, and modeling; understand the fundamentals of software design and architecture.
CO4	Discuss strategies for ensuring software quality, introduction to software testing methodologies, and understanding the importance of software security

Mapping of COs to PSOs

	PSO1	PSO2	PSO3	PSO4	PSO5
CO1	✓	✓	✓	✓	✓
CO2	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓

COURSE CONTENTS

Module 1: Introduction to Software Engineering, Nature of Software, Software Application Domains, Legacy Software (20 hours)

Module 2: Process Models, Agility and Agile Processes, Scrum, XP, Kanban, DevOps (20 hours)

Module 3: Understanding Requirements, Requirements Modeling, Design Concepts, Architectural Design, Component-Level Design, User Experience Design (20 hours)

Module 4: Quality Concepts, Software Reviews, Software Quality Assurance, Software Security Engineering, Software Testing at Component and Integration Levels (20 hours)

Module X (Teacher Specific): Quality Concepts, Software Reviews, Software Quality Assurance, Software Security Engineering, Software Testing at Component and Integration Levels (10 hours)

Core Compulsory Readings

1. Pressman RS Maxim BR. Software Engineering: A Practitioner's Approach. 9th ed. McGraw-Hill Higher Education; 2019.

Core Suggested Readings

1. Hans van Vliet, Software Engineering: Principles and Practice.
2. Robert C. Martin and Martin Fowler, Agile Software Development: Principles, Patterns, and Practices.

TEACHING LEARNING STRATEGIES

- Lecturing, Case study, Team Learning

MODE OF TRANSACTION

- Lecture, Seminar, Discussion, Questioning and Answering

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 4 credit courses (4 Credit theory).

Sample Questions to test Outcomes.

1. How do you handle software requirements changes during a project?
2. Can you describe the Agile software development methodology and its advantages?
3. What is the difference between unit testing and integration testing?
4. Explain concept of data flow diagram.
5. Give a description of the prototyping model.

KU5DSCCSE305 MACHINE LEARNING TECHNIQUES

Semester	Course Type	Course Level	Course Code	Credits	Total Hours
5	DSC	300	KU5DSCCSE305	4	90

Learning Approach (Hours/ Week)			Marks Distribution			Duration of ESE (Hours)
Lecture	Practical/ Internship	Tutorial	CE	ESE	Total	